

## Quiz: Quadratic Functions (2.3–2.4)

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Directions:** Show your work. Use exact values.

1. Rewrite in vertex form. Then state the vertex and the axis of symmetry.

$$f(x) = 2x^2 + 8x + 5.$$

2. For the quadratic function below, find the axis of symmetry and the vertex  $(h, k)$ .

$$f(x) = -3x^2 + 6x + 1.$$

3. For the equation below, compute the discriminant  $D = b^2 - 4ac$ , and state how many *real* solutions the equation has.

$$2x^2 + x + 1 = 0.$$

4. For the equation below:

(a) compute the discriminant and state how many *real* solutions there are,

(b) find all real solutions (exact form).

$$3x^2 - 5x + 1 = 0.$$